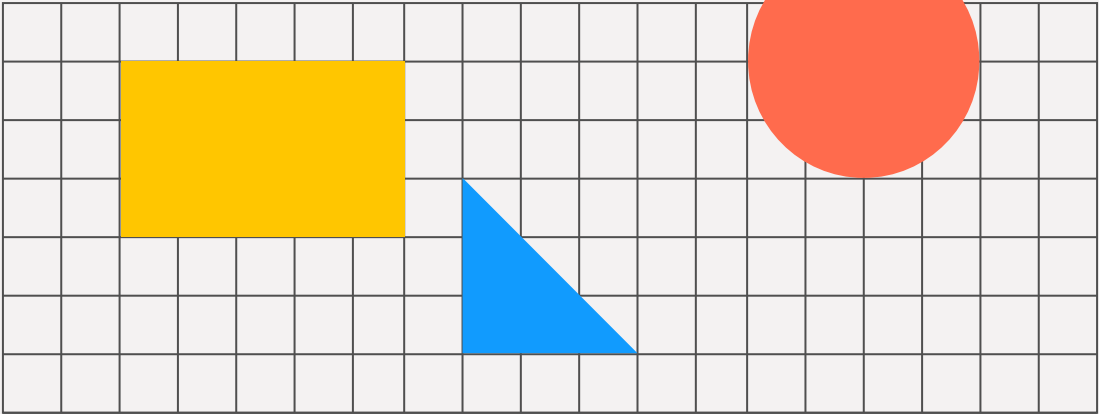


Prefix Sums

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The Scenic Route



Inspiration:

- Trans-Canada Highway

Figure:

- National Geographic

Illustration:

- Nano Banana Pro

The Scenic Route

Input/Output

Input Constraints

- $1 \leq N, M \leq 2\,000\,000$
- $-1\,000\,000 \leq V \leq 1\,000\,000$
- $1 \leq Q < 20\,000$
- $1 \leq K \leq 10$

Sample Input/Output

- Ordered
- Tied Case
- Impossible Case

Difficulty

- Kattis Medium



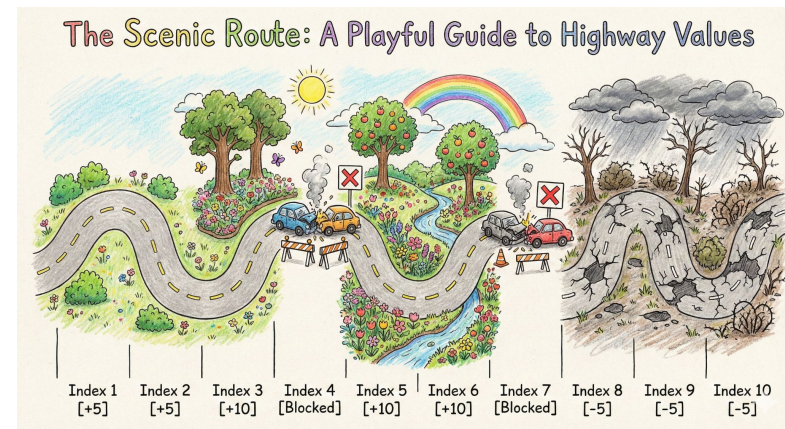
Test Cases

Sample Input

```
10 3
1 10 5
3 7 5
8 10 -10
3
4 7
5
1 2 3 4 5 6 7 8 9 10
```

Sample Output

```
20
30
Impossible
```



Correct, Efficient Solution

- Uses a difference array to apply M range updates in $O(1)$ each
- Converts the difference array into actual scenic values using a single prefix pass
- Builds a second prefix-sum array so any segment sum can be answered in $O(1)$
- Each query provides accident positions that block travel
- Sorts accident positions, then treats gaps between them as candidate drivable segments
- For each gap, computes scenic value using the prefix sums and tracks the maximum
- Outputs the best possible continuous segment; prints “Impossible” if none exist

Complexity: $O(M + N + Q \times K \log K)$

Incorrect Solution: Wrong Answer

Processing Accidents in Incorrect Order

- Accidents processed in input order instead of sorted order
- Result: Incorrect segment boundaries → wrong maximum scenic values
- For example, accidents [5, 2] creates barriers [0, 5, 2, 6] instead of [0, 2, 5, 6]
- Passes only test cases where accidents happen to be pre-sorted

Using Integer Instead of Long to Store Scenic Value

- Max scenic value can be $2,000,000 * 2,000,000 * 1,000,000 = 4e+18$
- Max Integer value $\sim 2e+9$
- Integer overflow occurs which results in wrap-around behaviour in Java and C++
- Requires array of type long

Incorrect Solution: TLE

- Direct array updates instead of difference array
- Creates new array slices for each segment
- Computes `sum()` directly instead of using prefix sums
- **TLE Complexity:** $O(M \times N + Q \times N)$
- Fails on large test cases ($N=2,000,000$, $M=2,000,000$, $Q=20,000$)

Incorrect Solution: Runtime Error

- Diff array size $N+1$ instead of $N+2$ → `IndexError` at `diff[R+1]` when $R=N$
- `prefix_sums` array size N instead of $N+1$ → `IndexError` accessing `prefix_sums[N]`
- Loop runs to N instead of $N+1$ → doesn't process final section
- Result: Array index out of bounds crashes on valid inputs

Test Case Generator

- Input Generation:
 - Creates m random beautification projects with random ranges $[left, right]$ and scenic value changes
 - Generates q traffic scenarios with varying numbers of accident locations
 - Ensures all constraints are satisfied
- Reproducibility:
 - Uses random seed (can be specified or auto-generated)
 - Allows deterministic test case recreation
- Usage: `python generator.py [n] [m] [q] [filename] [seed]`

Input Validator

- Validation Checks:
 - Verifies correct file structure (N M header, M project lines, Q header, Q scenario lines)
 - Ensures all values are valid integers
 - Checks data type consistency across all input lines
- Constraint Enforcement:
 - N: 1 to 2,000,000 (highway sections)
 - M: 0 to 2,000,000 (beautification projects)
 - Q: 1 to 20,000 (traffic scenarios)
 - Project ranges: $1 \leq L \leq R \leq N$
 - Scenic values: $-1,000,000 \leq V \leq 1,000,000$
 - Accident locations: $1 \leq a \leq N$, all distinct per scenario
- Usage: `python validate.py < [test file]`